Milestone 2

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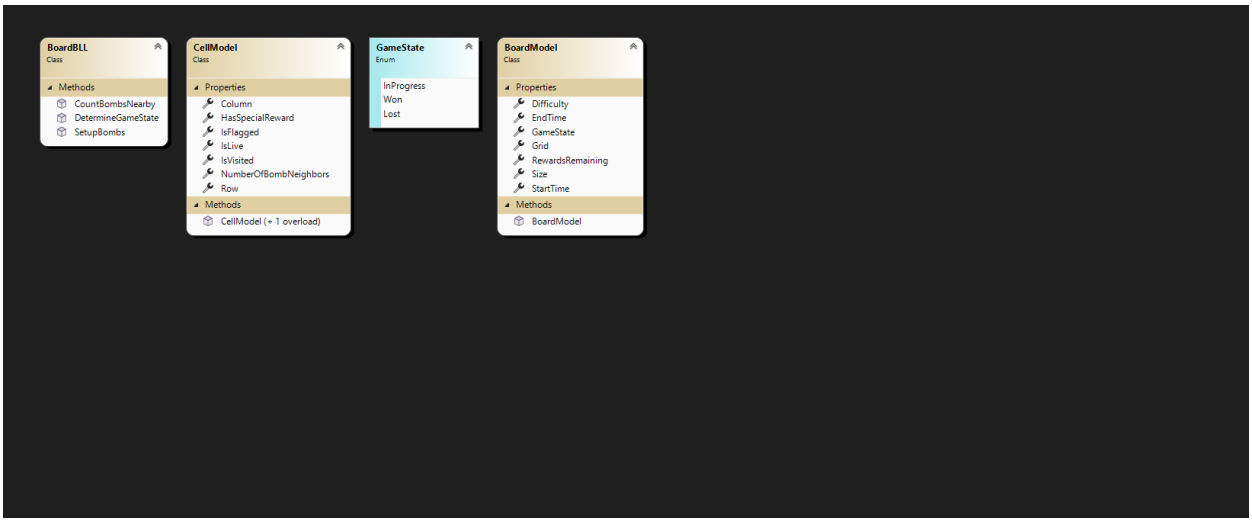
C# 2 - CST-250

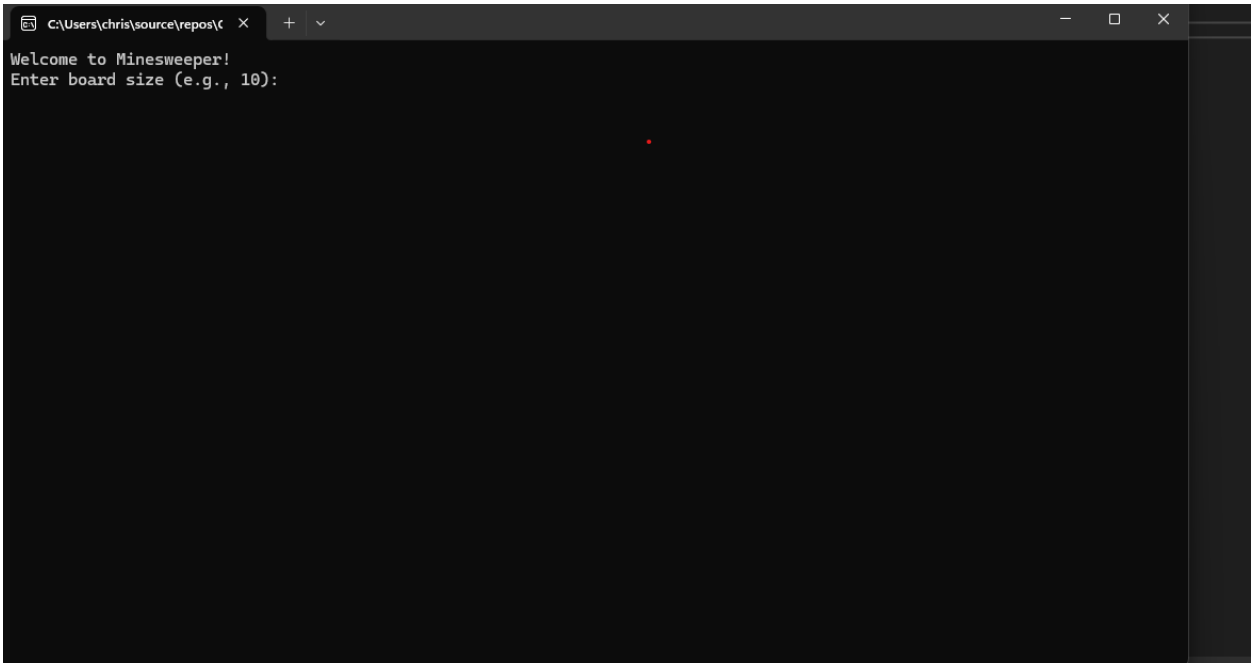
Make sure to copy the video link and paste to browser

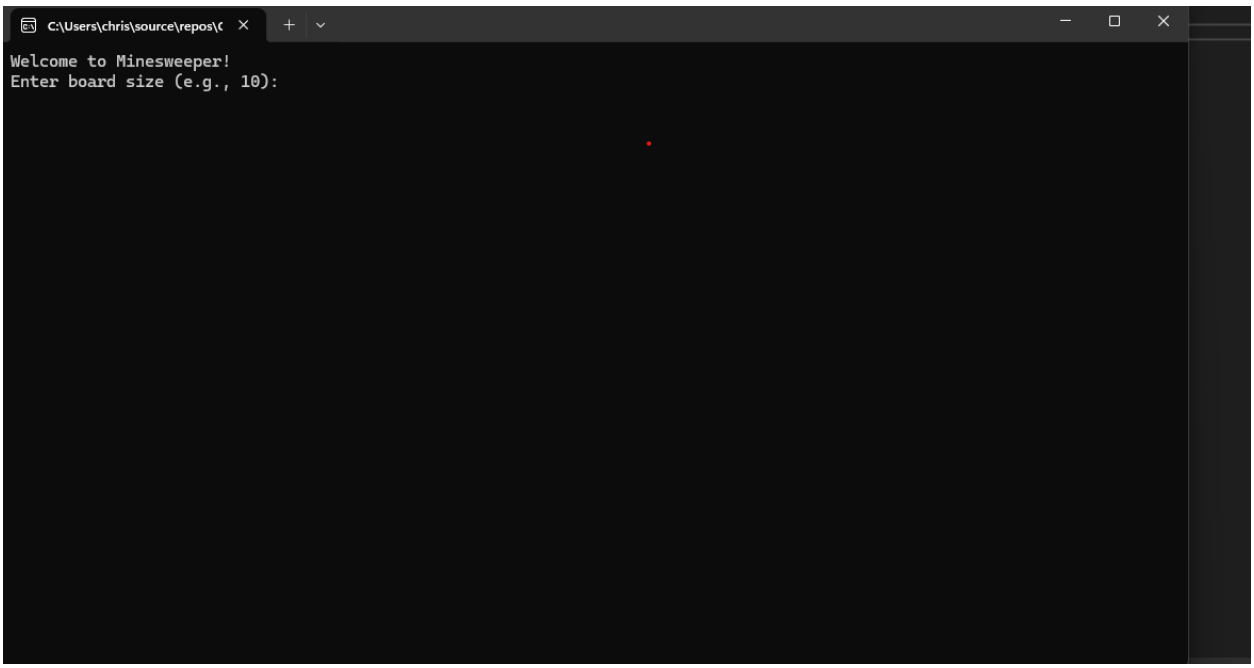
Github Link: https://github.com/chrisz99/MineSweeperClassLib

Screencast Video: https://drive.google.com/file/d/1SFAmGdahIGwNjtLvEe-ekU2500YyoYCN/view?usp=sharing

UML Diagrams







MineSweeper game running, prompting user for game board size.

```
C:\Users\chris\source\repos\C

    00 01 02 03 04 05 06 07 08 09
   -------------------------------
00 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
01 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
02 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
03 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
04 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
05 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
06 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
07 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
08 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
09 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |

Options: 1. Visit  2. Flag  3. Use Reward  4. Show Answers (Cheat)
Choose action and Row/Col (e.g., 1 5 5): 1 5 5
    00 01 02 03 04 05 06 07 08 09
   -------------------------------
00 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
01 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
02 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
03 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
04 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
05 | ?  ?  ?  ?  ?  .  ?  ?  ?  ? |
06 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
07 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
08 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
09 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |

Options: 1. Visit  2. Flag  3. Use Reward  4. Show Answers (Cheat)
Choose action and Row/Col (e.g., 1 5 5):
```

User initializes board of size of 10, user than checks for error handling, and also plays at the example slot 1 5 5.

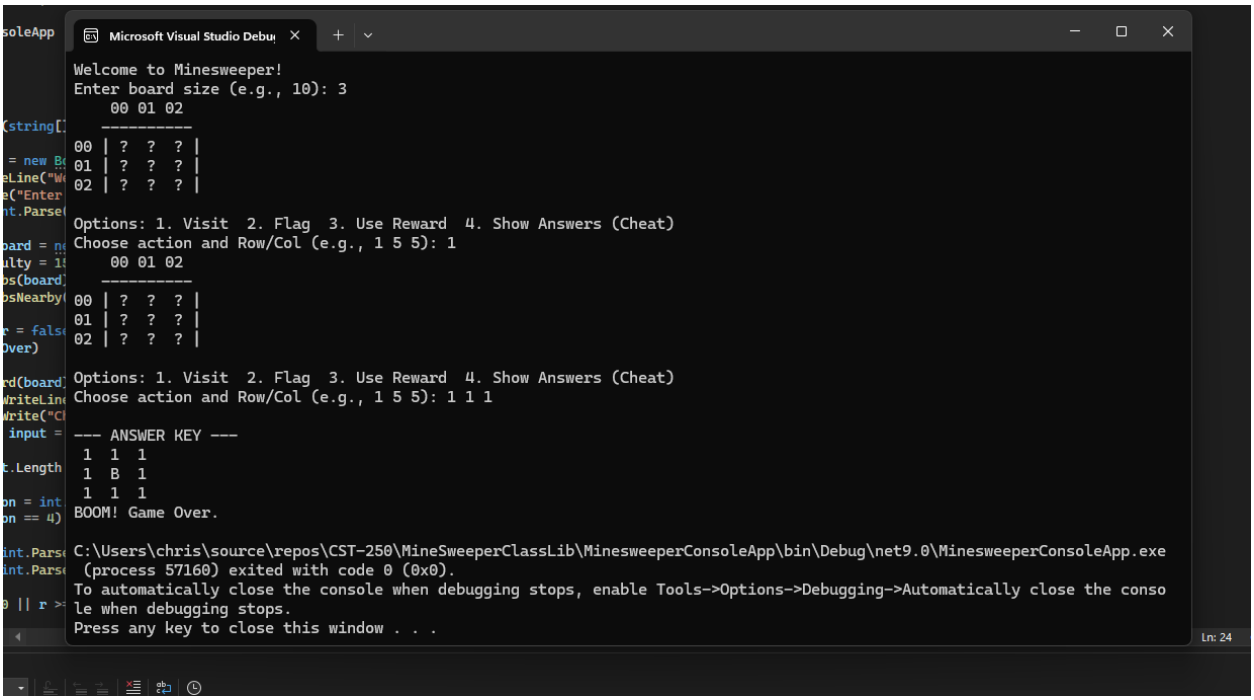
```
C:\Users\chris\source\repos\C

Options: 1. Visit  2. Flag  3. Use Reward  4. Show Answers (Cheat)
Choose action and Row/Col (e.g., 1 5 5): 4

--- ANSWER KEY ---
 0  0  1  B  3  2  1  0  1  B
 0  0  1  2  B  B  3  3  3  2
 2  2  1  1  2  3  B  B  B  1
 B  B  1  0  0  1  2  3  2  1
 4  4  3  1  1  0  0  0  0  0
 B  B  2  B  1  0  1  1  1  0
 3  3  3  1  1  0  1  B  1  0
 1  B  1  0  0  0  1  1  2  1
 1  1  1  0  0  0  0  0  1  B
 0  0  0  0  0  0  0  0  1  1
    00 01 02 03 04 05 06 07 08 09
   -------------------------------
00 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
01 | ?  F  ?  ?  ?  ?  ?  ?  ?  ? |
02 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
03 | ?  ?  ?  .  ?  ?  ?  ?  ?  ? |
04 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
05 | ?  ?  ?  ?  ?  .  ?  ?  ?  ? |
06 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
07 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
08 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |
09 | ?  ?  ?  ?  ?  ?  ?  ?  ?  ? |

Options: 1. Visit  2. Flag  3. Use Reward  4. Show Answers (Cheat)
Choose action and Row/Col (e.g., 1 5 5):
```

User plays some more, and also reveals answers.



User hits bomb and loses game

Summary

Milestone 2 transformed the static engine into an interactive experience by implementing a dynamic Game Loop that continuously listens for and processes user input. We introduced the Fog of War mechanic within our printing logic, ensuring squares remain hidden behind ? symbols until the IsVisited property is flipped to true. The Business Logic Layer was expanded with the DetermineGameState "Auditor," which scans the board after every move to automatically trigger Victory or Loss conditions. We implemented an Input Translator using int.Parse and Split to convert typed strings like "1 5 5" into actionable row and column coordinates. Finally, we integrated a Special Reward system that grants players a "Peek" ability, allowing them to neutralize risks by revealing a cell's hidden identity without stepping on it.